

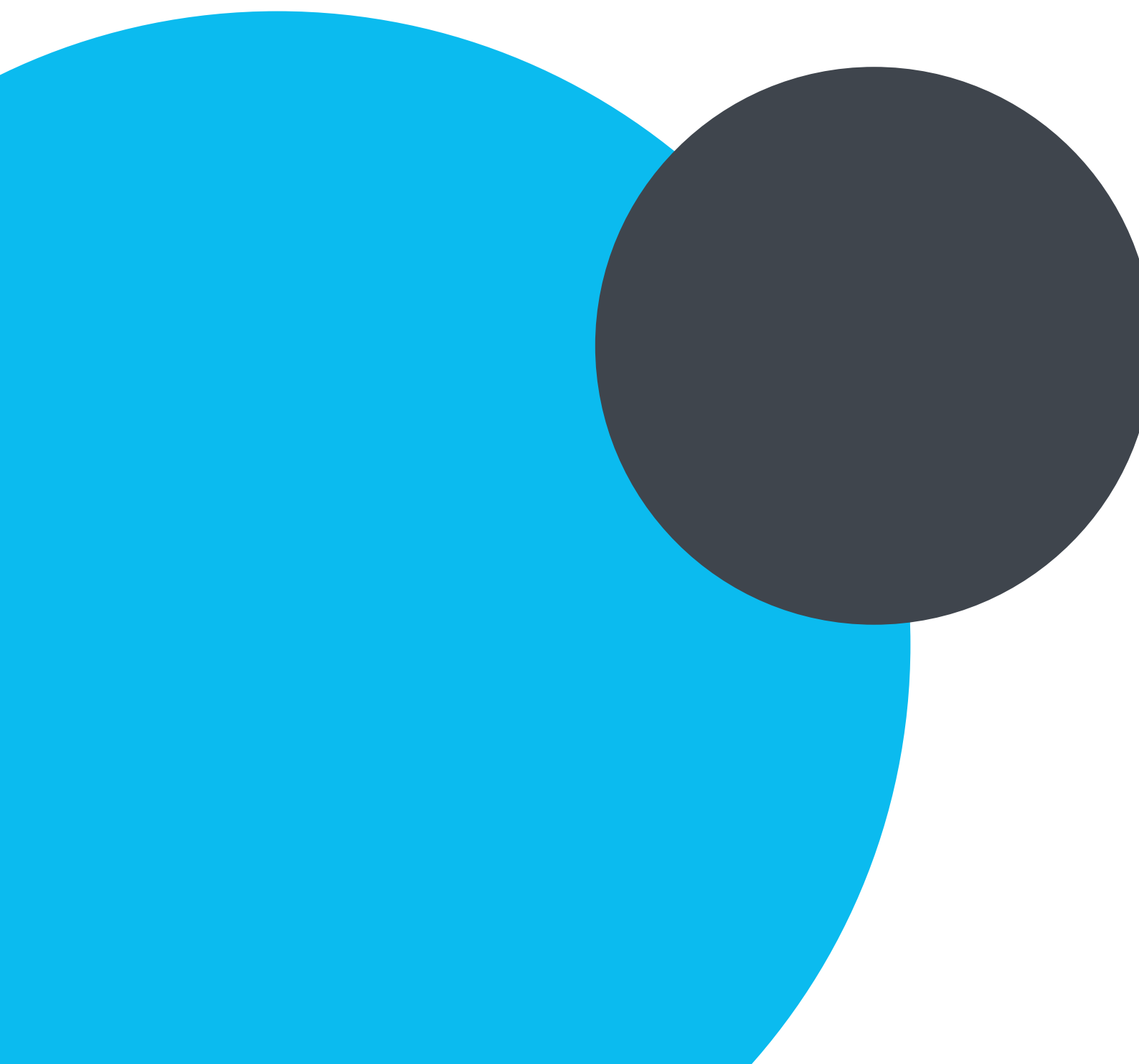
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Effect of stimulus duration on event-related potentials in a rapid serial visual presentation task

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Abstract

Previous research indicated that duration could be a confounding factor in EEG research. Therefore the effect of stimulus duration on event-related potentials (ERPs) was assessed in this study, using faces as visual stimuli. The study consisted of a rapid serial visual presentation (RSVP) task with log-uniformly distributed stimulus durations (100-1500 ms) and no inter-stimulus intervals (ISIs), as well as an identical control task with varying ISIs (800-2500 ms). Results showed no significant effect of stimulus duration on the P100, N170 and P300 ERP component peaks, but a significant positively correlated effect on the late stage of the P300 component (~300-500 ms latency). Further the study revealed a significant effect of ISIs on the P100, N170 and P300 components peak amplitudes. During a brief additional analysis an adaptation effect regarding the N170 component during a sequence of visual stimuli (face images) was observed. Finally a qualitative assessment of the results indicated that a combination of deconvolution-based overlap correction and stimulus duration modelling is methodologically sound.

Contents

Abstract	iii
Contents	v
List of Abbreviations	vii
1 Introduction	1
1.1 Event-related potentials	3
1.2 Effect of stimulus duration	5
1.3 Overlap effect	6
1.4 Goals of this study	7
2 Regression-based ERP estimation (rERP framework)	9
2.1 ERP averaging is equivalent to least-squares regression	10
2.2 Predictor coding schemes	11
2.2.1 Dummy coding	11
2.2.2 Treatment coding	12
2.2.3 Slope rERPs	12
2.2.4 rERP spline regression	13
2.3 Full-fledged regression models	14
2.3.1 Mass univariate model	14
2.3.2 Deconvolution model	16
3 Materials & Methods	19
3.1 Participants	19
3.2 Experimental setup	20
3.3 Experimental task	20
3.4 EEG Recording	21

3.5	Preprocessing	22
3.6	EEG analysis	23
3.6.1	Inter-stimulus interval (ISI) effect	24
3.6.2	Stimulus duration effect	24
3.6.3	Stimulus habituation effect	25
4	Results	27
4.1	Inter-stimulus interval (ISI) effect	28
4.2	Stimulus duration effect	31
4.3	Stimulus habituation effect	35
5	Discussion	37
5.1	Interpretation	37
5.1.1	Inter-stimulus interval (ISI) effect	37
5.1.2	Combination of overlap correction & duration modelling .	38
5.1.3	Stimulus duration effect	39
5.2	Limitations	40
5.3	Conclusion & Future Scope	41
	Bibliography	43
	Zusammenfassung	49
	Eigenständigkeitserklärung	51
	Usage of AI	53

List of Abbreviations

BCI Brain-computer interface

EEG Electroencephalography

EOG Electrooculography

ERP event-related potential

ISI inter-stimulus interval

MEG Magnetoencephalography

rERP regression-ERP

RSVP rapid serial visual presentation

SNR signal-to-noise ratio



Introduction

The brain is arguably the most complex, yet also mesmerizing organ of the human body. As the central organ of our nervous system it is responsible for coordinating almost all functions of our organism, ranging from cognitive and emotional over motoric functions, such as speech and muscle movements, to fully automatic processes, like breathing and our heart beat, which happen entirely unconscious. Even though researchers have made significant progress understanding our brain by identifying different regions and associating them with certain functions, our understanding of the system as a whole and how it enables extraordinary processes such as generating consciousness or producing thoughts and emotions, is still very limited. Despite this lack of wholistic understanding, scientist developed several techniques allowing us to observe activities in our brain, one of them being Electroencephalography (EEG).

EEG is a non-invasive neuroimaging technique that records electrical activity generated by the brain, through electrodes placed on the scalp. The electrodes enable to capture the voltage differences over time, providing a continuous measure of the brain's electrical dynamics with a high temporal resolution (Nunez & Srinivasan, 2006).

The primary sources of EEG signals are postsynaptic potentials generated by populations of cortical pyramidal neurons. When these neurons are synchronously activated, the summation of their electrical fields produce potential differences that can be measured by scalp electrodes, as illustrated in Figure 1.1. Because the skull and scalp act as volume conductors, EEG signals reflect the summed activity of thousands or millions of neurons rather than single-cell activity. This makes it particularly well suited for studying large-scale neural processes that unfold rapidly, such as sensory perception, attention and motor preparation (Luck, 2014; Nunez & Srinivasan, 2006).

EEG offers an excellent temporal resolution in the order of milliseconds, allowing researchers to examine the precise timing of cognitive processes. However, its spatial resolution is relatively low compared to other neuroimaging methods such as functional magnetic resonance imaging (fMRI), making it difficult to pinpoint the exact sources of measured activity. The recorded potentials represent a mixture of signals from multiple brain regions, and the inverse problem of determining the exact neural sources remains ill-posed (Grech et al., 2008; Nunez & Srinivasan, 2006).

Despite this limitation, its temporal precision and non-invasive nature have made EEG one of the most widely used tools in cognitive neuroscience, also making it highly suited for experimental paradigms where stimuli are presented in prompt succession.

One of these experimental paradigms which is highly relevant for the understanding of this thesis is the rapid serial visual presentation (RSVP) paradigm (Potter, 1976). Hereby visual stimuli are presented in rapid succession, with either none or very short inter-stimulus intervals (ISIs) - which are time between the end of a stimulus and the beginning of the next one - and stimulus durations in the hundreds of milliseconds. Typically there is a so called *target* stimulus, that differs in some form from the other stimuli - so called *distractors* - and the subjects are instructed to react to that target, in a given way (e.g. button press). The analysis of a task of this nature was the main focus of this work.

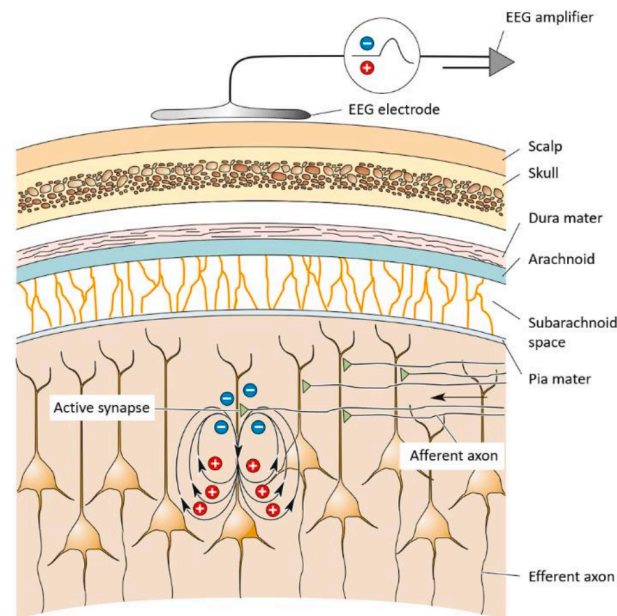


Figure 1.1 – Cortical pyramidal neurons produce postsynaptic potentials, generating electrical fields, which can be measured on the scalp via EEG electrodes. Changes in the membrane potential along a neuron, result in oppositely charged extracellular space, creating a dipole. Adapted from *Neurowissenschaften* (p.698) by Bear et al., 2018, Berlin Heidelberg: Springer, Copyright 2018 by Springer-Verlag.

1.1 Event-related potentials

In cognitive experiments, EEG data is often analyzed in the form of event-related potentials (ERPs), which are time-locked voltage fluctuations in the EEG that reflect neural responses to sensory, cognitive or motor events. Because the neural response to a single event contains substantial noise, elicited by other concurrent brain activities, ERPs are traditionally obtained by averaging the measured signal across many trials. This averaging enhances activity that is consistent in timing and polarity across trials while noise cancels out, uncovering the underlying ERP waveform (Luck, 2014).

ERP components are characteristic positive ('P') or negative ('N') deflections within this averaged waveform, named by their polarity and approximate time after event-onset (latency). Typical nomenclature for ERP components therefore is: *P100*, *N170* or *P300*. These three are all relevant components for

this work, which will be introduced later in this section. Figure 1.2 shows an exemplary ERP waveform from the analysis of this study, where all components are particularly well visible. A traditional distinction separates exogenous from endogenous components: exogenous components are mainly determined by physical stimulus properties and occur early (e.g., within the first 100 ms), whereas endogenous components are influenced by cognitive factors such as attention, task relevance or stimulus probability and typically appear at later latencies. This distinction however is more heuristic one rather than absolute, as many components show both sensory and cognitive modulation (Luck, 2014).

The *P100* is an early positive component peaking at ~100 ms latency. It is produced in the occipital area of the brain - precisely the early visual cortex - and is sensitive to visual stimulus features such as contrast and spatial location, making it a primarily exogenous sensory component (Russo et al., 2002).

The *N170* is a negative deflection peaking at around ~170 ms, strongest over posterior occipito-temporal sites, most likely elicited by the fusiform gyrus. The component is well known for its sensitivity to faces and face inversions and reflects relatively early perceptual processing of face structure (Bentin et al., 1996). Typically the component is larger in amplitude over electrodes in the right hemisphere (Rossion & Jacques, 2008) for faces as visual stimuli, emphasizing a strong involvement of the right occipito-temporal area in face perception.

The *P300* is a later positive component peaking roughly between 250–600 ms, strongly modulated by attention, stimulus probability and task relevance. Therefore it is seen as a classic endogenous component (Polich, 2007). It is likely produced by an interplay of various brain components, but can typically be measured most strongly over electrodes on the parietal lobe (Johnson, 1993).

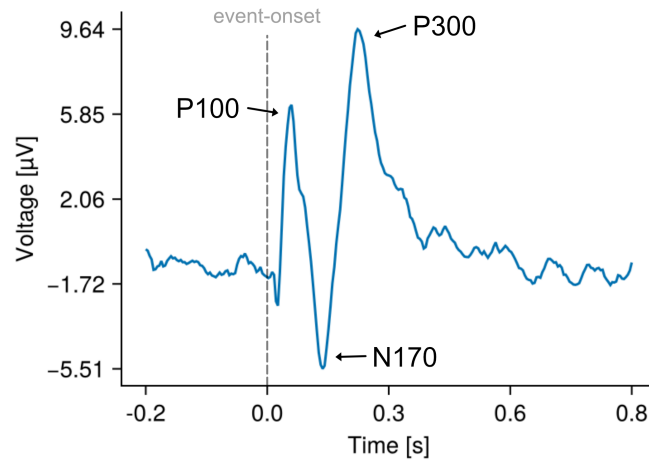


Figure 1.2 – Exemplary Event-related potential (ERP) waveform, to a face as visual stimulus. The face is presented at the event-onset marker. The signal was measured at the PO8 electrode in the right hemisphere of the occipito-temporal brain area.

Although components are convenient descriptors allowing a quantitative analysis of amplitude and latency, they do not necessarily correspond to single neural sources and their interpretation requires attention to potentially confounding influences such as the experimental design or scalp locality (Luck, 2014).

1.2 Effect of stimulus duration

The underlying assumption of ERP analysis is that for an event a fixed ERP waveform exists, which is distorted through noise caused by other neural processes. The traditionally used averaging technique hereby allows to recover that fixed ERP waveform, by taking the mean across many trials and therefore increasing the signal-to-noise ratio (SNR). However, previous research indicates that certain trial-wise influences exist, altering that supposedly fixed ERP. One of these trial-wise influences likely is the event duration, either caused through external stimulation (e.g. how long a stimulus is presented) or internal processes (e.g. how long a stimulus is processed by the brain) (Skukies et al., 2025). While the paper by Skukies et al. (2025) argues that especially the latter seems to be strong candidate for potential confounds, other research also indicates an effect of stimulus duration. A previous work by Tanskanen et al. (2007) showed in a Magnetoencephalography (MEG) study with face images as visual stimuli, that stimulus duration affected recognition performance but also

cortical responses. Specifically the M170 and M300 responses were affected, which are the MEG counterparts to the N170 and P300 ERP components.

While the combinations of RSVP tasks with ERP analysis has become a very popular approach - among other things for training Brain-computer interfaces (BCIs) to react to certain ERP components (Won et al., 2019) - the effect of stimulus durations in these paradigms is still mostly unexplored.

1.3 Overlap effect

Neural responses to sensory events can last from several hundred milliseconds to more than a second after stimulus onset. In experimental tasks where stimuli occur in rapid succession - such as RSVP paradigms - successive events might occur while the neural response to an event is still ongoing. As a result, the ERPs elicited by consecutive stimuli may temporally overlap, leading to additive or interfering effects in the recorded EEG signal, as illustrated in Figure 1.3 (Woldorff, 1993).

This temporal overlap poses a significant methodological challenge for traditional ERP averaging procedures, which assume that each event's neural response is isolated and temporally independent. When ISIs are short, overlapping activity from preceding or following stimuli can distort the measured waveform, complicating the interpretation of component amplitudes and latencies. The problem becomes particularly pronounced when adjacent stimuli share correlated properties — for example, in RSVP paradigms with alternating stimulus categories across consecutive presentations.

Despite this issue, short ISIs also offer experimental advantages. They allow researchers to investigate interactions between consecutive stimuli, such as priming or habituation effects and to collect a greater number of trials within a limited experimental duration. To mitigate the confounding effects of temporal overlap, one common approach is to introduce variability, or *jitter*, into the ISIs. This technique statistically reduces the contribution of adjacent events to the averaged ERP, helping to de-correlate overlapping neural responses (Luck, 2014).

More advanced correction methods have been developed to address overlap systematically. One notable example is the Adjacent Response (ADJAR) technique (Woldorff, 1993), which attempts to reconstruct and subtract the

contributions of temporally adjacent events through iterative approximation. However, ADJAR is a computationally complex method and relies on a series of manual adjustments, often resulting in uncertain accuracy. Consequently, more recent studies have turned to linear deconvolution approaches (see Chapter 2.3.2), which model overlapping ERPs within a regression framework to more precisely disentangle temporally adjacent neural responses.

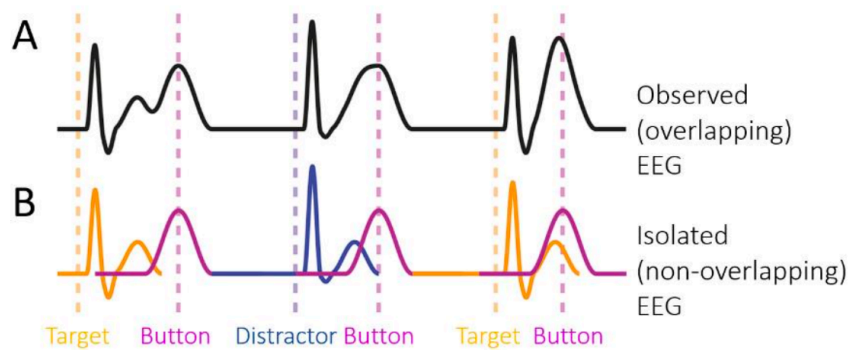


Figure 1.3 – Exemplary illustration of overlap effect on measured EEG signal. (A) Observed EEG signal for events with overlapping neural responses, causing observed EEG signal to be the linear sum of isolated responses to events. (B) Actual isolated responses to events without summation. From “Unfold: an integrated toolbox for overlap correction, nonlinear modelling, and regression-based EEG analysis” by B. V. Ehinger and O. Dimigen, 2019, PeerJ, 7, e7838. CC-BY 4.0.

1.4 Goals of this study

The present study aimed to investigate how the duration of visual stimuli influences neural responses in a RSVP task. Specifically, the experiment focused on faces as visual stimuli, assessing how varying stimulus durations modulate ERP components, associated with them - i.e. the P100, N170 and P300 components.

In addition to this scientific objective, the study pursued several methodological goals related to the analysis of overlapping neural responses in RSVP paradigms. First, it tried to evaluate the applicability of deconvolution-based overlap correction in combination with modelling of stimulus duration, within the regression-based ERP framework (see Chapter 2). The deconvolution technique hereby allows for the separation of temporally overlapping ERPs

by modeling each event's contribution to the continuous EEG signal, thereby providing a more accurate estimate of stimulus-locked neural activity. The modeling of stimulus duration on the other hand enables to assess the effect of varying stimulus duration on the ERP. However, due to the effect that the amount of overlap between neural responses and stimulus duration are highly correlated, the applicability of both in overlap correction and duration modeling in combination is still uncertain (Skukies et al., 2025).

Further, the study qualitatively compared linear and non-linear modeling approaches in order to quantify the relationship between stimulus duration and ERP waveforms. While linear modeling assumes a proportional relationship between duration and neural responses, non-linear modeling can capture more complex effects that may better reflect the underlying neurophysiological processes.

The study also examined the effect of ISIs on ERPs, by showing a 'blank' screen between stimuli, compared to a RSVP task (see Chapter 3.3 for detailed experimental description). These two differing task conditions were expected to provide insight into how ISIs and temporal jitter affect ERPs and model stability compared to a RSVP task.

Besides these main goals, a brief analysis regarding a possible habituation effect during a sequence of consecutive visual stimuli was conducted, in order to provide further explanations regarding the results of the main analysis.



Regression-based ERP estimation (rERP framework)

The underlying assumption of ERP analysis is that for a given stimulus - in the context of this work, the presentation of a face image - the brain produces a fixed pattern of neural activity. Due to the fact that there are many neural processes going on simultaneously, it's not as simple as just presenting a stimulus and measuring the brain response, since there will be a lot of noise elicited by other processes that distort the measured signal. The traditional approach of calculating an ERP therefore would be to average the measured brain response across several trials, which should cancel out the noise, allowing to recover the actual ERP waveform. While this ERP averaging method has been incredibly successful in the field, it's shortcomings are that it can only be applied to a limited range of experimental designs. To overcome these drawbacks the regression-based *rERP framework* (Smith & Kutas, 2015a; 2015b) was introduced, allowing ERP averaging to handle an arbitrary combination of categorical and continuous covariates, partial confounding, non-linear effects and overlap correction between distinct events, all within a single unified model. In this section the foundations of the *rERP framework*, according to the work by Smith & Kutas (2015a; 2015b) will be explained. Their framework also

constitutes the ground work for the Unfold.jl toolbox (Ehinger & Alday, 2025), used for the ERP analysis in this thesis.

2.1 ERP averaging is equivalent to least-squares regression

In order to estimate the ERP to an event, we need to calculate the ERP value at one single electrode and time point after event-onset (latency) and then repeat this procedure across all latencies to uncover the actual ERP. This means we're trying to estimate a single value β , being the ERP value for a specific electrode and latency (e.g. electrode PO8, 150ms post-event). In order to estimate this value β we need the measurements across several trials for this electrode and latency. These measurements across n trials we refer to as $y_1, y_2, \dots, y_n$ and we assume that each of these measurements consists of the summation of the actual ERP value β , as well as a individual noise term for each trial. Therefore we can write the formula for an arbitrary trial i :

$$y_i = \beta + \text{noise}_i \quad (2.1)$$

Since we don't know the noise values for any of the trials we can't simply solve for β . Here the least-squares principle can help us, which says we should choose our β value so that it minimizes the total squared noise:

$$\text{squared noise} = \sum_{i=1}^n (\text{noise}_i)^2 = \sum_{i=1}^n (y_i - \beta)^2 \quad (2.2)$$

In order to do that we first take the derivative of the squared noise (Equation (2.2)) and set it to 0, which enables us to find a local minimum. This gives us the following formula:

$$\begin{aligned} -2 \left(\sum_{i=1}^n y_i - \sum_{i=1}^n \beta \right) &= 0 \\ \Leftrightarrow \beta &= \frac{1}{n} \sum_{i=1}^n y_i \end{aligned} \quad (2.3)$$

Equation (2.3) shows that minimizing the squared noise, according to the least-squares principle, is essentially the same as estimating β by taking the mean of our measured values $y_1, y_2, \dots, y_n$. Therefore the least-square regression approach is mathematically equivalent to the traditional ERP averaging.

2.2 Predictor coding schemes

In the previous section we have shown that the least-squares regression produces equivalent estimates for the ERP than the traditional ERP averaging technique. However, in order to enable models to handle combinations of covariates we need to extend our equation to a general least-squares regression formula:

$$y_i = \beta_1 x_{i,1} + \beta_2 x_{i,2} + \dots + \text{noise}_i \quad (2.4)$$

By doing this we can integrate an arbitrary number of covariates as predictors $x_{i,j}$ into one regression model. In order to this, several coding schemes for categorical predictors exist, which will be explained in the following.

2.2.1 Dummy coding

The dummy coding scheme allows to estimate the ERP for several categorical values of a predictor at once in the most straightforward way. If we consider the categorical covariate *blank* (whether there was a ‘blank’ screen before the face image) from our study, we would define our predictors with dummy coding as following:

$$x_{i,1} := \begin{cases} 1, & \text{if blank} = \text{true} \\ 0, & \text{if blank} = \text{false} \end{cases} \quad x_{i,2} := \begin{cases} 0, & \text{if blank} = \text{true} \\ 1, & \text{if blank} = \text{false} \end{cases}$$

Applying this coding to the following regression formula:

$$y_i = \beta_1 x_{i,1} + \beta_2 x_{i,2} + \text{noise}_i \quad (2.5)$$

the least-squares fitting would result in β_1 as the average of the *blank* trials and β_2 as the average the *no blank* trials. Therefore we are able to capture both ERP estimates inside one model.

2.2.2 Treatment coding

Another commonly used method of handling categorical variables in the rERP framework is by treatment coding. In this coding scheme all except one of the categorical values are coded in the dummy coding scheme. The one categorical value that isn't dummy-coded is referred to as the reference level or intercept term. Applying the treatment coding scheme to the same example as used for the dummy coding, with the *blank* trials as our reference level, results in the following predictors:

$$x_{i,1} := 1 \quad x_{i,2} := \begin{cases} 0, & \text{if blank} = \text{true} \\ 1, & \text{if blank} = \text{false} \end{cases}$$

Applying this coding scheme to the Equation (2.5) would result in β_1 being the average of the *blank* trials and β_2 the difference ERP between the *no blank* and *blank* trials. The name treatment coding originates from exactly this property enabling a effortless comparison between various 'treatments' to a control condition, like in our case the *blank* condition.

2.2.3 Slope rERPs

The coding schemes introduced so far focussed on categorical predictors and even though they are quite useful by enabling us to estimate several ERP waveforms within one model, they didn't bring in any new functionalities compared to the traditional ERP averaging. Slope rERPs do that, by allowing to directly analyze the linear effect of continuous covariates. In order to showcase their functionality we will explain them on the example of stimulus duration, which was a continuous covariate and also one of the main focuses of this thesis. Assuming the stimulus durations are ranging from 100ms to 1500ms we we define our codings for the predictors as follows:

$$x_{i,1} := 1 \quad x_{i,2} := \text{stimulus duration for trial } i$$

Once again applying this coding scheme to the Equation (2.5), the β_1 value can be interpreted as baseline intercept term for a minimal stimulus duration of 0ms. β_2 estimates the change of the ERP value with a positive unit change of the stimulus duration. One unit would in this example be 1ms, considering we use milliseconds as our unit scale. This enables us to estimate the ERP value for an arbitrary stimulus duration value, by simply altering the $x_{i,2}$ term. One

could for example estimate the value for a stimulus duration of 200ms with the following formula:

$$\text{EEG (200ms stimulus duration)} = \beta_1 \times 1 + \beta_2 \times 200 \quad (2.6)$$

Since β_2 is the slope regression line that relates $x_{i,2}$ and y_i we refer to it as a slope rERP. Also notice how this definition of predictors is conceptually quite similar to the *treatment coding*. $x_{i,1}$ also denotes an intercept term and its corresponding parameter β_1 gives a baseline ERP value for the stimulus duration 0. The predictor $x_{i,2}$ then codes for derivations from this baseline, based on the stimulus duration.

2.2.4 rERP spline regression

Slope rERPs introduced in the previous section enable us to analyze the effect of continuous covariates, however they come with the restriction that only linear relationships between the covariate and the measured signal can be modelled. While in many cases the simplifying assumption of a linear model might in fact be able to capture general directionality of a relationship, there are situations where it's particularly valuable to give the regression model the freedom to account for non-linear relationships in the data. One of these situations is when examining the nature of the relationship is itself a target of the analysis, like it is the case in this work. One approach to integrate non-linear models into our least-squares regression framework is by using a set of cubic b-splines as basis functions $f_{1(x)}, f_{2(x)}, \dots, f_{k(x)}$, as described in the accompanying paper to the introduction to the rERP framework (Smith & Kutas, 2015b). By taking the weighted sum of the spline functions:

$$\alpha_1 f_{1(x)} + \alpha_2 f_{2(x)} + \dots + \alpha_k f_{k(x)} \quad (2.7)$$

we are able to construct wide variety of non-linear curves. In order to integrate them into the framework model, one must choose a set of basis functions and allow the regression models to find the parameters a_i which fit the data best. Figure 2.1 illustrates an example of how rERP spline regression is applied. This approach is also used in the Unfold.jl toolbox (Ehinger & Alday, 2025) to model non-linear relationships.

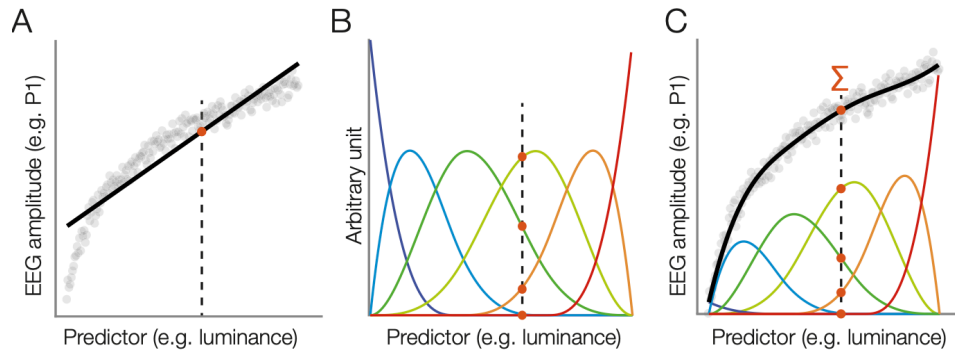


Figure 2.1 – Modeling a non-linear relationship with a set of spline functions. **(A)** Example of a linear modeling approach for a relationship between a predictor (e.g. stimulus luminance) and a dependent variable (e.g. EEG amplitude). The linear function does not fit the data well. **(B)** Set of six overlapping b-spline functions, distributed across the range of the predictor. This generates six new predictor values, of which only three are non-zero for the chosen predictor value (dashed vertical line) in this example. **(C)** Each spline $f_{k(x)}$ is weighted with its estimated predictor value α_k . For the prediction of the dependent variable (here EEG amplitude) the spline function values (red dots) for the given predictor value are summed up. Because the splines are overlapping, this produces a smooth, non-linear fit to the data. From “Unfold: an integrated toolbox for overlap correction, nonlinear modelling, and regression-based EEG analysis” by B. V. Ehinger and O. Dimigen, 2019, PeerJ, 7, e7838. CC-BY 4.0.

2.3 Full-fledged regression models

Combining predictors for categorical and continuous covariates - defined according to the coding schemes introduced in the previous section - within the same regression formula allows to account for all of them inside one model. Two distinct modelling approaches exist hereby: the first one called the *Mass univariate model*, where no overlap correction between the ERPs of consecutive events is conducted, and the second one being the *Deconvolution model*, where overlap correction is applied using linear deconvolution. Both of these approaches will be explained in the following.

2.3.1 Mass univariate model

In the first step of the mass-univariate approach (Pernet et al., 2011) the EEG data is cut into epochs according to a time window around the onsets of the

events. One could choose for example a time window of 200ms before the stimulus onset until 800ms after. For each time point t in the epoch we then fit a separate regression model, which tries to estimate the corresponding ERP value. If we consider an exemplary model with one categorical covariate of two category values (A, B), encoded with the treatment coding scheme, we get the following formula:

$$\text{EEG}_{\text{epoch},t} = b_{1,t} \times 1 + b_{2,t} \times \text{covariate} = B + e_{\text{epoch},t} \quad (2.8)$$

In this case the categorical value A would be the intercept term or control condition and covariate=B an indicator, with values 0 or 1, whether the covariate has the categorical value B. $e_{\text{epoch},t}$ represents the noise term for the epoch at hand at time point t . This Equation (2.8) can now be written in matrix notation for all epochs as:

$$\text{EEG}_t = Xb_t + e_t \quad (2.9)$$

X is the so called design matrix. In this case, where each row represents one epoch and therefore one event. b_t is a vector of the estimated β , which need to be interpreted according to the used coding scheme. In our example b_t would consist of the ERP value for category A and the difference ERP value between category B and A, both at time point t , since we used the treatment coding scheme. If we now run this model again for each time t in the epoch, we get the entire regression-ERP (rERP). Figure 2.2 illustrates the explained process once again in a graphic manner.

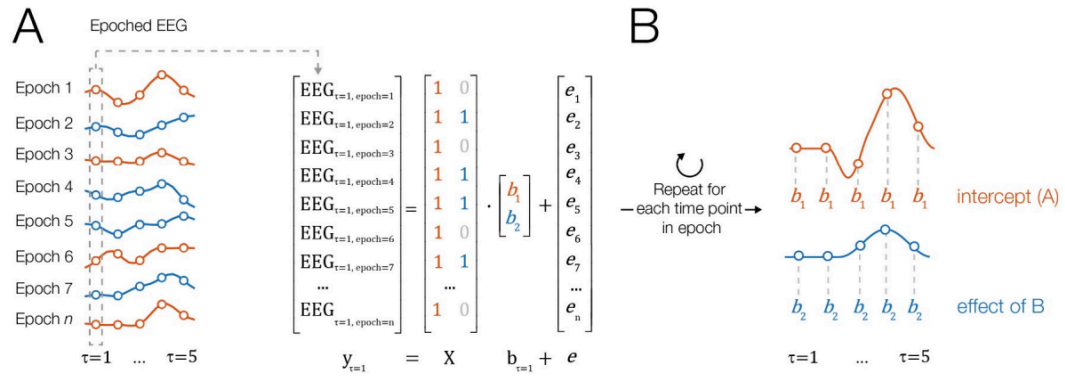


Figure 2.2 – Illustration of the mass univariate model. **(A)** Visualization of the regression model for time point $\tau = 1$ **(B)** Resulting rERP waveforms, according to the used coding scheme (here treatment coding), obtained by running regression model, for each time point τ in the epoch. From “Regression-based analysis of combined EEG and eye-tracking data: Theory and applications” by O. Dimigen and B. V. Ehinger, 2021, Journal of Vision, 21(1), 3. CC-BY-NC-ND 4.0.

The approach of the mass univariate model can be extended to more complex regression models incorporating several categorical and continuous covariates, by simply expanding the design matrix, so that it contains more predictors. However, it cannot account for varying temporal overlap between neighboring events, causing it to produce biased estimates if overlap differs between conditions (Dimigen & Ehinger, 2021).

2.3.2 Deconvolution model

The deconvolution method has been adapted from the analysis of the hemodynamic response in functional magnetic resonance imaging (fMRI) (Dale & Buckner, 1997). The linear deconvolution approach directly addresses the problem of overlapping brain responses (Dimigen & Ehinger, 2021). Figure 2.3 shows an illustration of an exemplary deconvolution model, with two conditions A and B. Initially we have the continuous EEG data, shown on the left side of sub-figure A of Figure 2.3. The measured EEG values for given samples (i.e. time points in recording) are then used as the y values in our regression formula. Further the design matrix is expanded according to the chosen window length, which specifies the time window after an event that is affected by its elicited neural response. In the illustrated example model, events which could be of condition A or B were estimated to have an effect for five time points post

event-onset, which can be comprehended by the diagonals of 1's in the design matrix following an event-onset. The model hereby essentially uses the dummy coding scheme indicating whether an event has an effect on a specific sample. The model further contains a separate noise e_{sample} for each sample. This model design results in the parameters b_t estimating the ERP values for all time points t , inside the chosen time window and therefore the whole ERP waveform for the different conditions. Sub-figure B displays the regression formula for time point 25, which shows that the model assumes the measured EEG value of that sample is composed of the neural responses to two condition A events (b_1, b_5) and one condition B event (b_9).

The deconvolution model approach can as well be extended to incorporate several categorical and continuous covariates within one model and the resulting rERP waveforms can be analyzed just like traditional ERPs (Dimigen & Ehinger, 2021). However, there is to say that the linear deconvolution approach can only work, if ISIs vary for stimulus sequences. Otherwise there is a colinearity effect and no method will be able to reliably disentangle the neural responses (Smith & Kutas, 2015b).

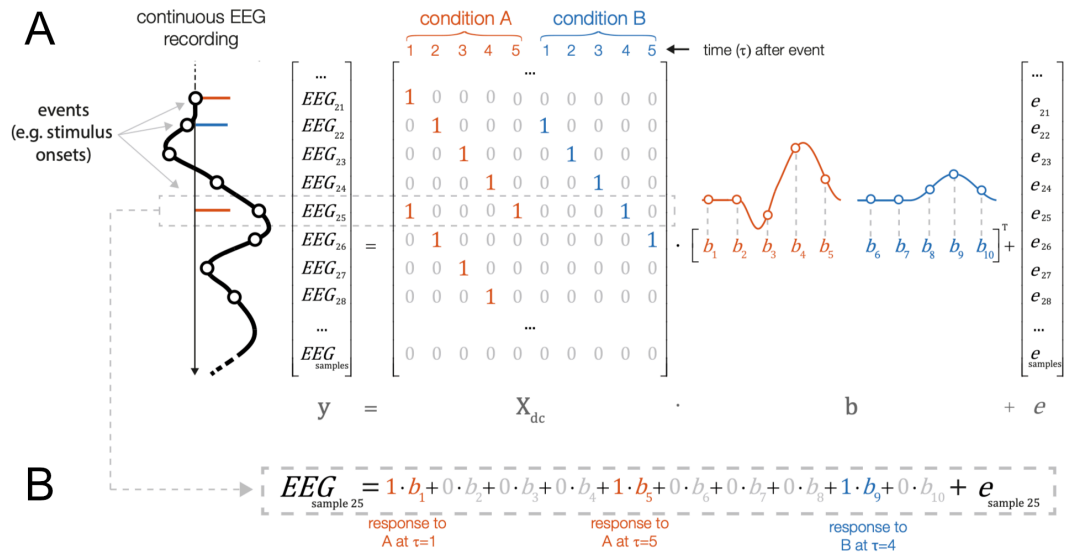


Figure 2.3 – Illustration of an exemplary deconvolution model. **(A)** Visualization of the regression model components. It is indicated that the regression parameters b_t capture the estimated ERP waveform at sample 25 in the EEG recording, which is the onset time of a condition A event, is highlighted as an example. **(B)** Regression formula belonging to highlighted sample 25. The formula highlights that the measured EEG value at sample 25 is composed of neural response to two condition A events (b_1, b_5) and one condition B event (b_9), plus a noise term ($e_{\text{sample 25}}$). From “Unfold: an integrated toolbox for overlap correction, nonlinear modelling, and regression-based EEG analysis” by B. V. Ehinger and O. Dimigen, 2019, PeerJ, 7, e7838. CC-BY 4.0.

/ 3

Materials & Methods

The EEG data analyzed in this work was collected during a study conducted by Martin Geiger as part of his master's thesis (Geiger, 2022). The description of the study is mainly based on his work. In the following I will specify the materials of the study - which was approved by the local ethics committee of the University of Stuttgart - as well as the methods it was analyzed with.

3.1 Participants

The study consisted of 38 right-handed participants (14 female, 24 male) from 21 to 36 years of age (mean: 27.2, SD: 3.5). 26 of the participants had normal and 12 corrected-to-normal vision, all passing an eyesight test, confirming 6/6 vision prior to the experiment. Additionally they were verbally queried for exclusion criteria, such as tinnitus, photosensitive migraine or epilepsy, as well as other severe neurological or psychiatric disorders. They had to give written informed consent and received a monetary compensation of 15€/hour for the participation in the study.

3.2 Experimental setup

The participants were seated alone in a dimly lit room on a rotation- and translation-locked chair. For stimulus duration a LED display (ACER Nitro XV252QF) was placed at a viewing distance of approximately 90cm, which was controlled via a PC (HP EliteDesk 600 G6, GPU: AMD Radeon RX 550X). Stimuli were presented with Psychtoolbox-3 (Brainard, 1997; Kleiner et al., 2007; Pelli, 1997) via MATLAB 2022a (The MathWorks Inc.). Participants had to respond to distractors with a button press on a response box (MilliKey SR-5 r1), placed on the table in front of them.

3.3 Experimental task

Participants had to complete two tasks in randomized order during the course of the study: one visual oddball task and one RSVP task. Since this work centers around the RSVP task and the analysis of the oddball task was already conducted in Martin Geiger's master's thesis (Geiger, 2022), this section solely elaborates on the design of RSVP task.

During the RSVP task each participants had to complete 6 blocks of trials in total. 3 of the blocks were of the *no-blank* condition and 3 of the *blank* condition. The order of the blocks was hereby at random.

Each block of the *no-blank* condition consisted of a series of 120 face images which were presented in the middle of the screen in rapid succession. The duration for how long a face image was presented was sampled from a log-uniform distribution ranging from 100 to 1500 ms. Additionally with every face image there was a 10% probability that a red dot would appear in the middle of the face, as a distractor, to which the participants were instructed to react as quickly as possible, with a button press. The distractor was always visible for a fixed duration of 100 ms and there was at least a gap of 4 s between consecutive distractors. Figure 3.1 A also illustrates the description the *no-blank* condition.

Each block of the *blank* condition had an identical structure regarding the face image presentation and distractors compared to the *no-blank* condition. The only difference between the two conditions was that for the *blank* condition a 'blank' screen containing only a fixation cross in the middle was shown for a duration between 800 to 2500 ms between face images. The duration was hereby also sampled from a log-uniform distribution. The *blank* condition trials

should hereby function as a control group for which the overlap between stimulus responses is reduced compared to the *no-blank* condition trials, due to the ‘blank’ screens in-between. A log-uniform distribution was chosen for the stimulus duration and ISI duration in order to keep the hazard rate (Herbst et al., 2018; Nobre et al., 2007) - which is the instantaneous probability that an event (here the stimulus onset/offset) will occur - equal at each time point of the task. An illustration of the described *blank* condition is shown in Figure 3.1 B.

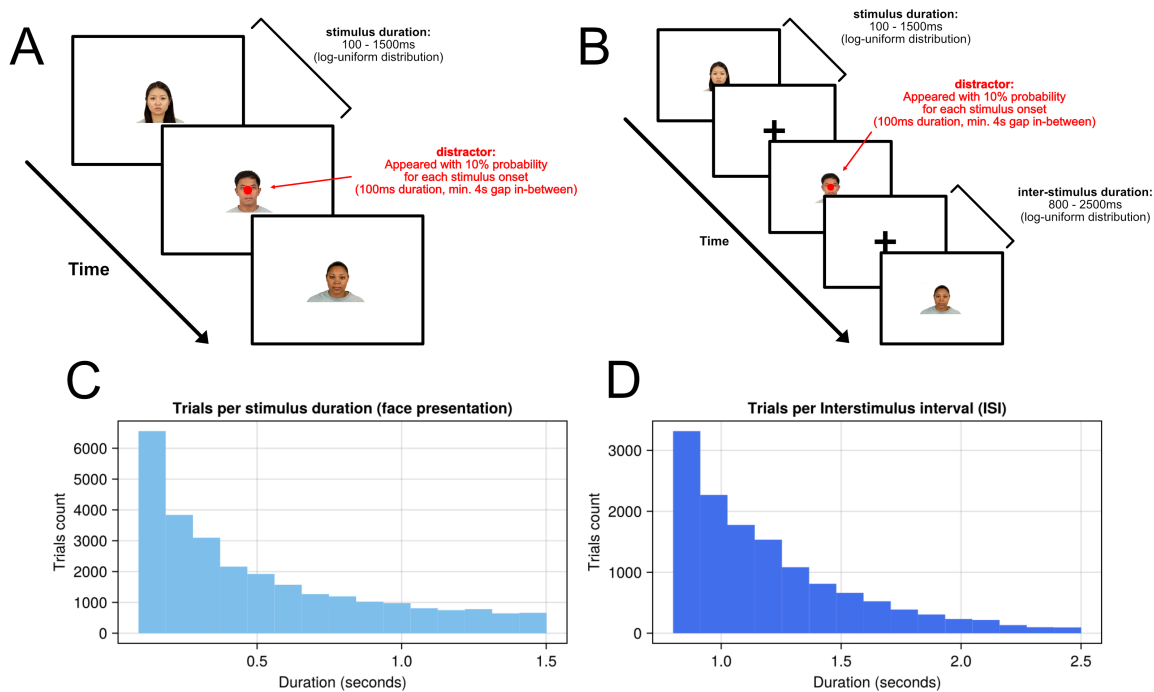


Figure 3.1 – (A) Schematic description of the of the RSVP task, with no inter-stimulus intervals (ISIs), referred to as the *no-blank* condition. (B) Schema of the task, with varying ISIs (log-uniformly distributed), referred to as the *blank* condition. (C) Histogram of stimulus duration (i.e. presentation time of a face image), across all trials from both the *blank* and *no-blank* condition. (D) Histogram of ISIs, across all trials the *blank* condition.

3.4 EEG Recording

For the recording of continuous EEG and Electrooculography (EOG) data an AntNeuro (Enschede, Netherlands) System comprised of an eegoTM sports amplifier (EE-225) and waveguardTM original cap (CA-203.s1) was used. A

128-channel cap was placed according to the 10/5 system and data was recorded from 60 Ag/AgCl scalp electrodes, whose locations are visualized in Figure 3.2. Additionally two horizontal EOG ring electrodes were placed lateral to the outer canthi of the left (HEOGL) and right (HEOGR) eye and two vertical EOG electrodes above (VEOGU) and below (VEOGL) the midline of the left eye. The ground (GND) electrode was placed at the left earlobe, and CPz electrode was used as an online reference and the data was digitized at a sampling rate of 1000 Hz.



Figure 3.2 – Channel locations of all recorded EEG electrodes, plotted on a scalp topoplot. The electrodes of particular interest are the ones in the occipito-temporal area of the scalp (i.e. P7, PO7, P8, PO8, O1, O2), since neural responses to visual stimuli, including faces, can typically measured the strongest in this area.

3.5 Preprocessing

The data was preprocessed in MATLAB 2021a (The MathWorks Inc.), with a custom script based on the paper by Pernet et al. (2021). First channel locations were added and the EEG signals were downsampled to 250Hz. Then 50Hz line noise elicited by electrical lines in Europe was removed, using Zapline-plus (Cheveigné, 2020; Klug & Kloosterman, 2022). Channels with a flatline for longer than 5 s, a correlation of less than 0.8 to the rest of the data and line noise greater than 4 standard deviations, were removed. Then data was re-referenced to the average of all remaining channels and a copy of the data was made. Large spikes where the absolute signal amplitude exceeds 1000 μ

V were removed and a 1.5Hz lowpass filter was applied to the copied data. Subsequently the adaptive mixture independent component analysis (AMICA) (Palmer et al., 2011) was run on the copy of the data for 1000 iterations. After, the ICLabel algorithm (Pion-Tonachini et al., 2019) was applied to the results. This algorithm generated probabilities for estimated components, which are essentially origins of the measured EEG signal. These components were classified as Brain, Muscle, Eye, Heart, Line Noise, Channel noise or Other. All non-brain components with classification probability of $>80\%$ were removed and weights, producing this effect, were applied to the original EEG data. Then artefact subspace reconstruction (ASR) (Kothe, 2013) was used to mark temporally short, high-variance signals in the continuous data, which were later removed from the data before analysis. Finally, bad channels were interpolated again using the signals of neighboring channels and data was high-pass filtered at 0.1 Hz.

3.6 EEG analysis

Since neural responses to faces as visual stimuli were analyzed, the main focus of the analysis was on the ERP components P100, N170 and P300, which were already introduced in Chapter 1.1. According to previous literature, these components can typically all be measured strongly in the the occipito-temporal area of the scalp (Johnson, 1993; Rössion & Jacques, 2008; Russo et al., 2002), therefore the P7, PO7, PO8, P8, O1, O2 electrodes were chosen as the channels for the ERP analysis. A time window from -200ms to 800ms around the stimulus onset was used for the ERP analysis, as well as for the stimulus effect in the deconvolution models.

The preprocessed EEG data was analyzed using the *Unfold.jl* toolbox (Ehinger & Alday, 2025) coded in the *julia* programming language (Bezanson et al., 2017). This toolbox allows to generate rERP regression models as described in Chapter 2. As a notebook-like environment the Pluto.jl (Van-der Plas, 2025) was used in order to create interactive analysis scripts.

At the beginning of the analysis event files were extended so that every face stimulus event was marked whether it was of the *blank* or *no-blank* condition and which number of stimulus it was in the sequence of the current block, which was later used to investigate a possible habituation effect during a block sequence. Then, faulty segments marked by the ASR algorithm in the preprocessing were removed from the data. Two subjects (subjects 8, 25) were excluded from

further analysis after individual ERP inspection of all subjects, because their ERPs were too noisy to be considered in the analysis. Therefore 36 subjects remained.

3.6.1 Inter-stimulus interval (ISI) effect

In order to assess the difference of the ERPs from the RSVP task, referred to as the *no-blank* condition, and the task with varying ISIs (*blank* condition), a regression model was fitted with the following Wilkinson formula, which is common notation to describe regression models:

$$\text{ERP} \sim 1 + \text{blank} \quad (3.1)$$

This model only considered the categorical covariate *blank*, indicating the condition of an event. The 1 term indicates the usage of the treatment coding scheme (Chapter 2.2.2) with the *no-blank* condition as the intercept. The model design was then implemented as a mass univariate and a deconvolution model, in order to compare the two modelling approaches.

Finally baseline correction with the mean of a 200 ms time window pre-stimulus was conducted and a Cluster-depth permutation test (Frossard & Renaud, 2022) was applied across all relevant channels (P7, PO7, PO8, P8, O1, O2) to identify regions in the ERPs where there is a statistically significant difference between the two conditions. The Cluster-depth permutation test ($\alpha = 0.05$) hereby used 5000 permutations and a cluster-forming threshold based on the 95th percentile of the t-distribution.

3.6.2 Stimulus duration effect

For the analysis of the stimulus duration effect, separate models were defined to model the relationship in a linear and non-linear manner. For the linear model the following regression model, described in Wilkinson notation was used:

$$\text{ERP} \sim 1 + \text{duration} \quad (3.2)$$

Here the model used the rERP slope technique (Chapter 2.2.3), modeling the effect of the continuous covariate *duration* linearly.

The non-linear models had the following Wilkinson formula:

$$\text{ERP} \sim 1 + \text{spl}(\text{duration}, 5) \quad (3.3)$$

There the effect was modeled in a non-linear way, by using rERP spline regression (Chapter 2.2.4), with 5 B-splines as basis functions. The values for the covariate *duration* hereby ranged from 100 to 1500 ms, since the presentation time for a face image was randomly sampled from a log-uniform distribution over this time frame, as described in the experimental task description (Chapter 3.3).

Furthermore the duration effect was modeled separately for the *blank* and *no-blank* condition, since the relationship might differ between the two conditions. The linear and non-linear effect modeling approaches were then applied in a mass univariate and a deconvolution model, which were then ran on the EEG data of the individual subjects and thereon a grand averaged ERP across all subjects was calculated. Finally the marginal effect of the continuous covariate stimulus duration was calculated for the different models, indicating the effect of the covariate on the rERP waveform. This marginal effect was then tested for statistical significance, using the Hotelling's T^2 test (Hotelling, 1931), with a significance level $\alpha = 0.05$. The test essentially checks if at least one model coefficient belonging to the covariate stimulus duration is not 0 at each latency, therefore testing whether a stimulus duration effect exists.

3.6.3 Stimulus habituation effect

In order to investigate a possible habituation effect, the effect of the covariate indicating the number of a stimulus in the current block - or equivalently, how many stimuli were already presented beforehand in the current block - was assessed. This covariate could adopt a value between 1 and 120, since one trial block always consisted of 120 face images as visual stimuli. As for the duration effect a linear and non-linear modeling approach, described by the following Wilkinson formulas was implemented. For the linear model, using slope rERPs:

$$\text{ERP} \sim 1 + \# \text{ stimulus in sequence} \quad (3.4)$$

For the non-linear model, using spline regression with 5 B-splines:

$$\text{ERP} \sim 1 + \text{spl}(\# \text{ stimulus in sequence}, 5) \quad (3.5)$$

Both were then realized as a mass univariate and deconvolution model.

For this effect no statistical tests were conducted, since it was only an additional analysis, which arose from findings during the main analysis in order to better understand the findings.

/4

Results

The results of the EEG analysis are presented in three sections, beginning with an assessment of the inter-stimulus interval (ISI) effect. This allows to get a general understanding of the difference in neural responses between the *blank* and *no-blank* task condition, before discussing further effects. Second, the results regarding the stimulus duration effect will be discussed, which was the main focus of this work. Since the stimulus duration effect was modeled individually for the *blank* and *no-blank* condition, there will also be a distinction between the two conditions for the stimulus duration effect. Finally a brief section will present the findings regarding the habituation effect during a sequence of visual stimuli.

4.1 Inter-stimulus interval (ISI) effect

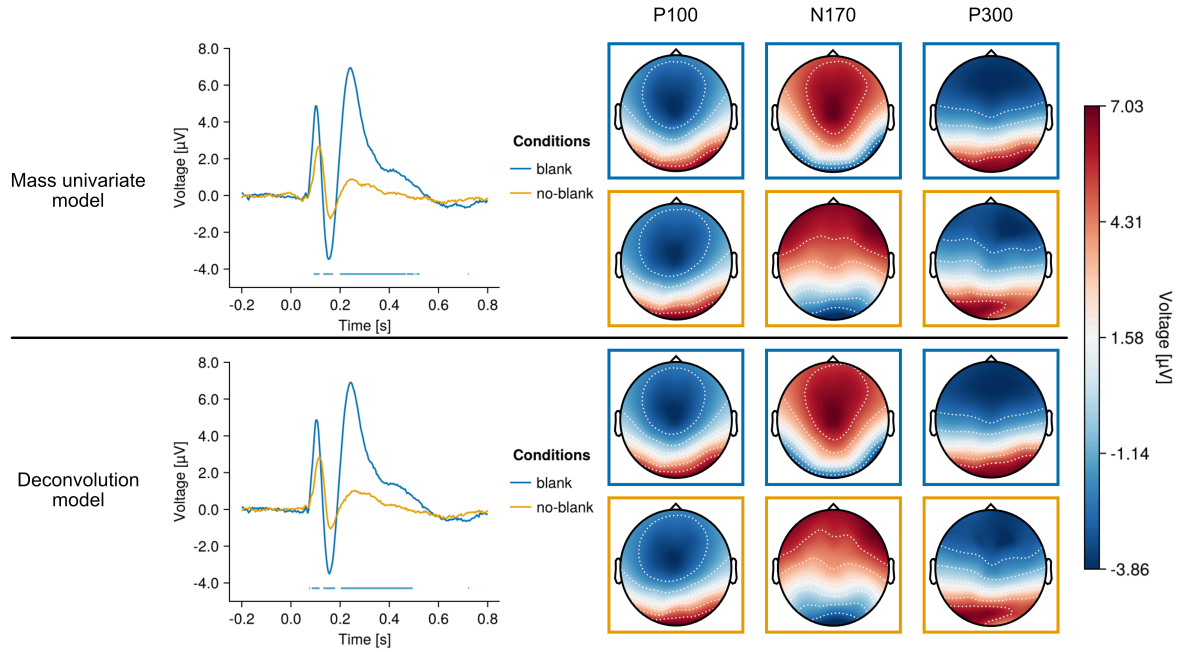


Figure 4.1 – Grand averaged ERPs, for the RSVP task (*no-blank* condition) and task with varying inter-stimulus intervals (*blank* condition) at PO8 electrode. Baseline correction according to the mean signal pre stimulus (-200ms to 0ms time window) was applied. The lines underneath the ERP plots depict the regions where a statistically significant difference between the two conditions was determined by the cluster-depth permutation test. Additionally topoplots for each ERP component (P100, N170, P300), at latency of peak amplitude of grand averaged ERP are displayed for both conditions, indicated by the border color of the topoplots. The rows further distinct the mass univariate and deconvolution models.

The ERPs of the *blank* and *no-blank* conditions differed significantly for the P100, N170 and P300 ERP components in both the mass univariate and deconvolution model, as shown by the significant regions of the ERP plots in the Figure 4.1, which were determined by the cluster-depth permutation test. Hereby the regions of statistical significance were almost identical between the two modelling approaches. The grand averaged ERP waveforms for channel PO8 as well as the general measured neural activity, visualized in the topoplots was also very similar for both modelling approaches.

Mod- els	Mean peak latency (ms)					
	P100		N170		P300	
	no- blank	blank	no- blank	blank	no- blank	blank
Mass univari- ate	107 (25)	105 (12)	178 (44)	157 (29)	277 (50)	242 (19)
Decon- volu- tion	106 (28)	106 (12)	179 (44)	157 (29)	281 (54)	245 (20)

Table 4.1 – Overview of the latencies of ERP components peaks averaged across all subjects, measured at the PO8 electrode. The latency values are the mean time points post-event where the specific ERP component reaches its maximum amplitude. For the P100 and P300 this is where the maximum μ V value is reached in the ERP, since they constitute positive potentials. For the N170 component it's where the minimum μ V value, since it's a negative potential. The values in brackets are the corresponding standard deviations to the mean latencies. The latency estimates of both modeling approaches (mass-univariate or deconvolution) and both task conditions, the RSVP task (*no-blank* condition) and the task with varying ISIs (*blank* condition), are given.

The mean latencies of the component peaks (Table 4.1), which were averaged across the ERPs of all subjects, were virtually identical for both modelling approaches. However, there was a difference between conditions especially regarding the N170, P300 components. For both of them the peak latency was higher in the *no-blank* trials, indicating a delayed onset of the components, compared to the *blank* condition. Standard deviations of the mean latencies were generally higher in the *no-blank* condition trials for all three ERP components.

Mod-els	Mean peak amplitude (μV)					
	P100		N170		P300	
	no-blank	blank	no-blank	blank	no-blank	blank
Mass univariate	3.39 (1.97)	6.08 (3.60)	-2.07 (1.44)	-5.12 (3.70)	1.66 (1.10)	7.70 (3.57)
Deconvolution	3.57 (2.12)	6.11 (3.62)	-1.97 (1.53)	-5.15 (3.67)	1.86 (1.32)	7.65 (3.53)

Table 4.2 – Overview of the peak amplitudes of the ERP components averaged across all subjects, measured at the PO8 electrode. The amplitude values are the mean maximum/minimum voltage values of the corresponding ERP component. They essentially are the corresponding ERP values to the latencies values from Table 4.1. For the P100 and P300 this is the maximum μV value of the components, since they constitute positive potentials. For the N170 component it's the minimum μV value, since it's a negative potential. The values in brackets are the corresponding standard deviations to the mean amplitudes. The amplitude estimates of both modeling approaches (mass-univariate or deconvolution) and task conditions, the RSVP task (*no-blank* condition) and the task with ISIs (*blank* condition), are given.

The mean amplitudes of the component peaks (Table 4.2), which were averaged across the ERPs of all subjects, were as well very similar between the two models, with differences in the hundredths of a μV . However, similar to the grand averaged ERP the amplitudes differed significantly between the *blank* and *no-blank* condition. Generally the amplitudes of all three components were much higher for the *blank* condition. Especially the endogenous P300 component had a very strong peak in the *blank* condition, where as for the *no-blank* condition the peak almost completely vanished. The standard deviations the mean amplitudes were consistently higher in the *blank* condition, inverting the observation for the standard deviations of the mean peak latencies.

4.2 Stimulus duration effect

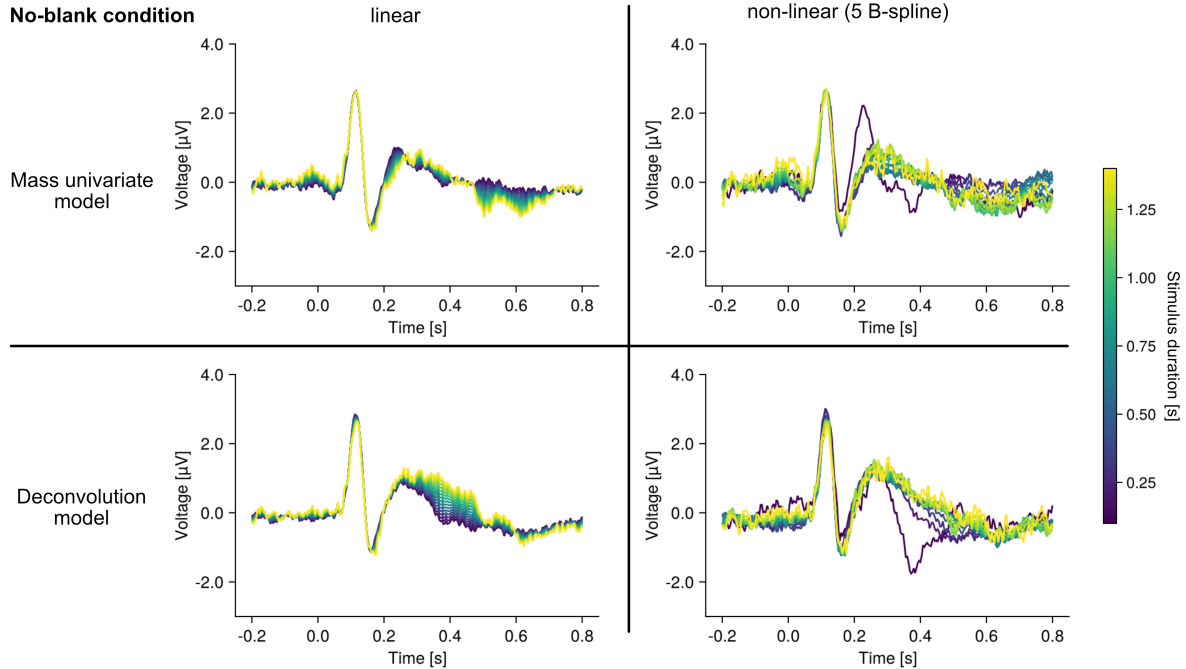


Figure 4.2 – Grand averaged ERPs (PO8 channel) for the RSVP task (*no-blank* condition) modelled at varying stimulus durations, indicated by the colorbar on the right. The row differs between different modelling approaches (mass univariate, deconvolution) and the column between linear and non-linear models.

For the RSVP task (*no-blank* condition) the stimulus duration didn't have any significant effect on the P100 and N170 components in any of the models, as can be seen in the ERPs of Figure 4.2. However for the later stage of the P300 component (~300-500 ms post stimulus onset) there seemed to be an effect in the deconvolution models, where a longer stimulus duration elicited a stronger measured electrical potential. Results from the Hotelling's T^2 also showed a statistically significant effect in the occipito-temporal region in this time window for these models, showing a statistically significant effect from ~200-400 ms in the non-linear deconvolution model and ~300-400 ms latency in the linear deconvolution model (Figure 4.4). This effect however was much stronger pronounced in the non-linear deconvolution model, resulting in an almost negative peak at ~380ms for the shortest stimulus duration. The non-linear effect at latency 380ms is also depicted in Figure 4.5. In the non-linear

mass-univariate model there is furthermore a very suspicious peak before the P300 component for very short stimulus durations. This effect however could not be statistically verified with the statistical test.

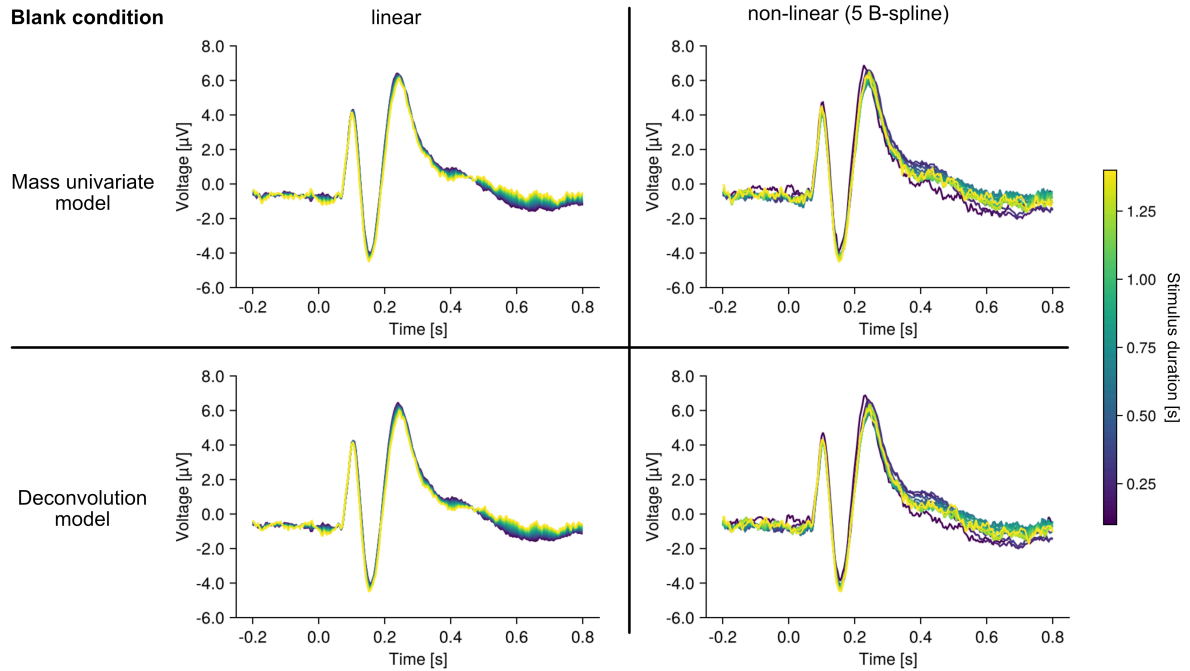
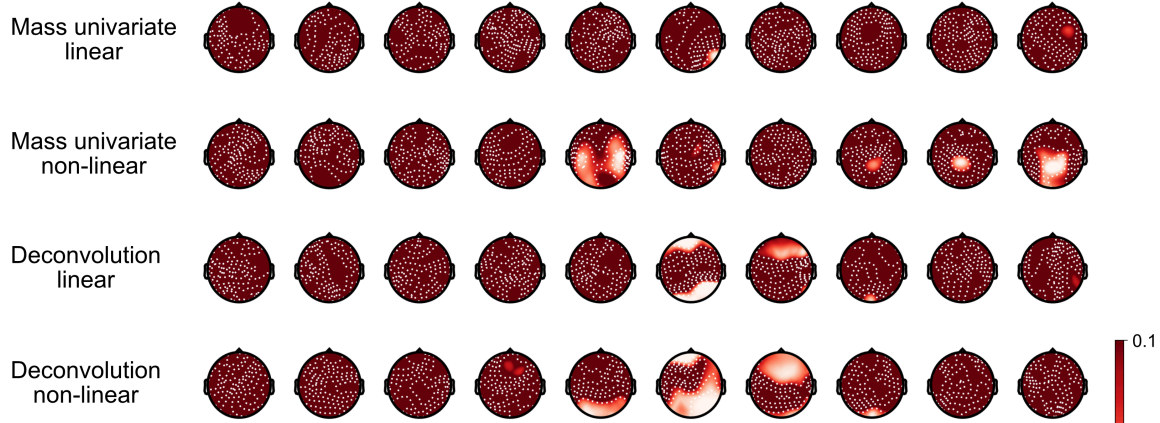


Figure 4.3 – Grand averaged ERPs (PO8 channel) for the task with varying inter-stimulus intervals (*blank* condition) modelled at varying stimulus durations, indicated by the colorbar on the right. The figure contains different modelling approaches (mass univariate, deconvolution), as well as linear and non-linear models.

For the task with varying ISIs (*blank* condition) the models estimated a much less visible effect of the stimulus duration, compared to the *no-blank* condition (Figure 4.3). Results of the Hotelling's T^2 test showed no statistical significance for any of the ERP components peaks in the occipito-temporal regions. Though there was again a significant effect in the late stage of the P300 component in the occipito-temporal region, specifically time window of ~400-500 ms latency for the non-linear mass univariate model and ~400-600 ms for the non-linear deconvolution model (Figure 4.4). This effect however is very hard to retrace in the ERP plots of Figure 4.3, since the effect is not monotonous (see Figure 4.5 *blank* condition, 380ms latency). Further the non-linear deconvolution model

also showed significant activity in the parietal region from ~200-600 ms latency according to the Hotelling's T^2 test p-values.

No-blank condition



Blank condition

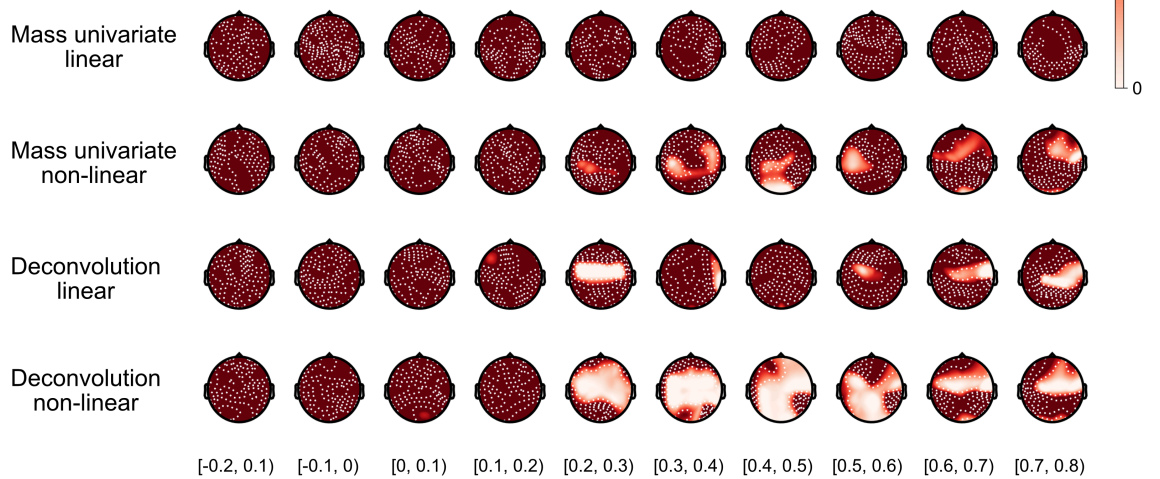


Figure 4.4 – Topoplots series' with p-values from the Hotelling's T^2 test for the two task conditions (*blank* and *no-blank*) as well as all models. The dots on the topoplots refer to the recorded EEG channels and the coloring represents their corresponding p-values according to the colorbar. With the chosen significance level being 5%, light red to white coloring indicates a significant stimulus duration effect for the channel in the given time interval, which are shown at the bottom of the figure.

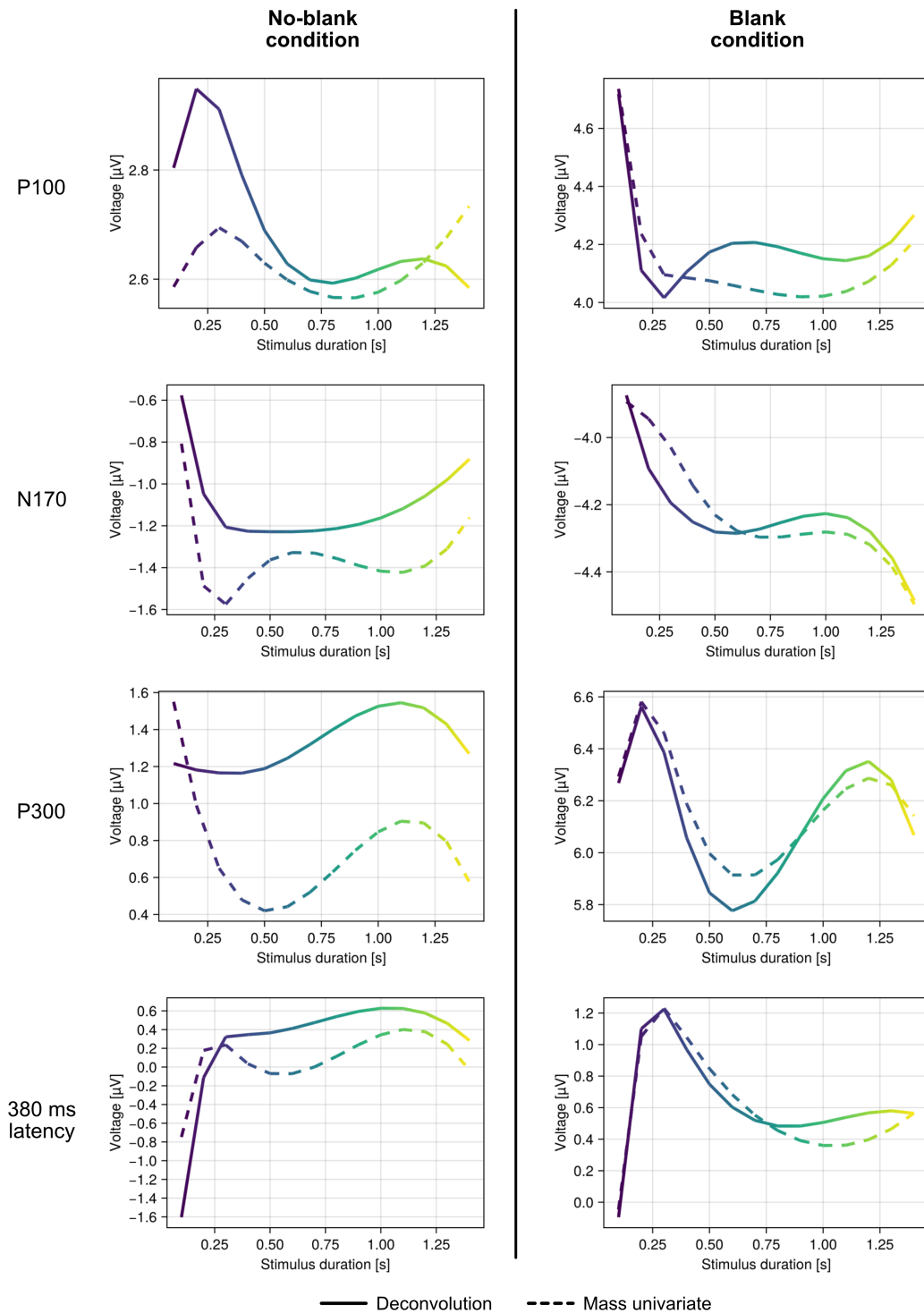


Figure 4.5 – Estimated non-linear duration effect on peak amplitudes of ERP components, measured at PO8 channel, for both task conditions. The effect is plotted for latency of the grand averaged ERPs component peaks. The figure only shows the effects of the non-linear models, which used 5 B-splines as basis functions. The last row also shows the effect at 400 ms latency, being the time point where duration effects are statistically significant in some models.

Figure 4.5 visualizes the estimated duration effect of the non-linear models at the latency of the ERP component peaks. For the *blank* conditions the effects are estimated almost identical by the mass univariate and deconvolution model, with only slight deviations.

In the *no-blank* conditions however there is a clear difference in the shape of the estimated relationship, especially for the P100 and P300 component. Nevertheless the voltage differences between the models never exceed $0.8 \mu\text{V}$ for the component peaks.

4.3 Stimulus habituation effect

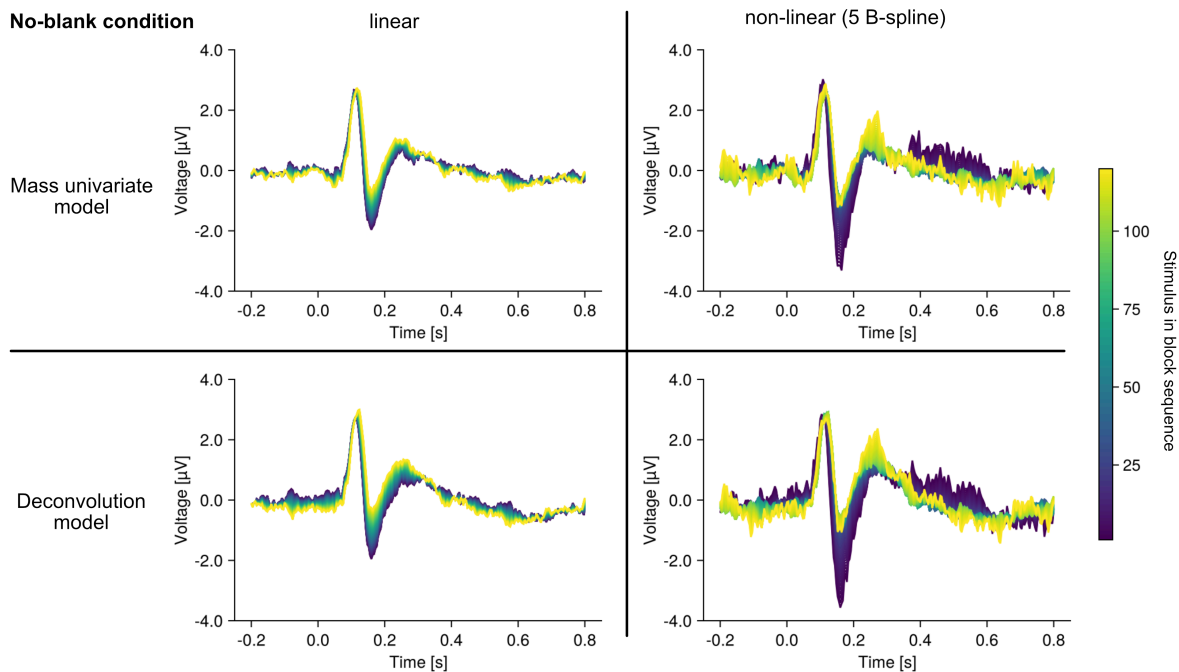


Figure 4.6 – Grand averaged ERPs (PO8 channel) for the RSVP task (*no-blank* condition) modelled at different points in stimulus sequence of a trial block, indicated by the color bar on the right. The row differs between different modelling approaches (mass univariate, deconvolution), and the column between linear and non-linear (5 B-spline) models.

For the RSVP task (*no-blank* condition) where 120 stimuli in a block were presented without any ISIs between them, the estimated ERP waveforms of all models showed some effects (see Figure 4.6). Especially the amplitude of the

N170 component seemed to decrease strongly within the stimulus sequence, indicating a habituation effect. For the P300 component however there seemed to be an increase of peak amplitude within a stimulus sequence, in both non-linear models, but also in the linear deconvolution model. The non-linear models both also seemed to show increase with longer stimulus sequence for the later part of the P300 component. The P100 component generally seemed unaffected during the stimulus sequence in the *no-blank* condition.

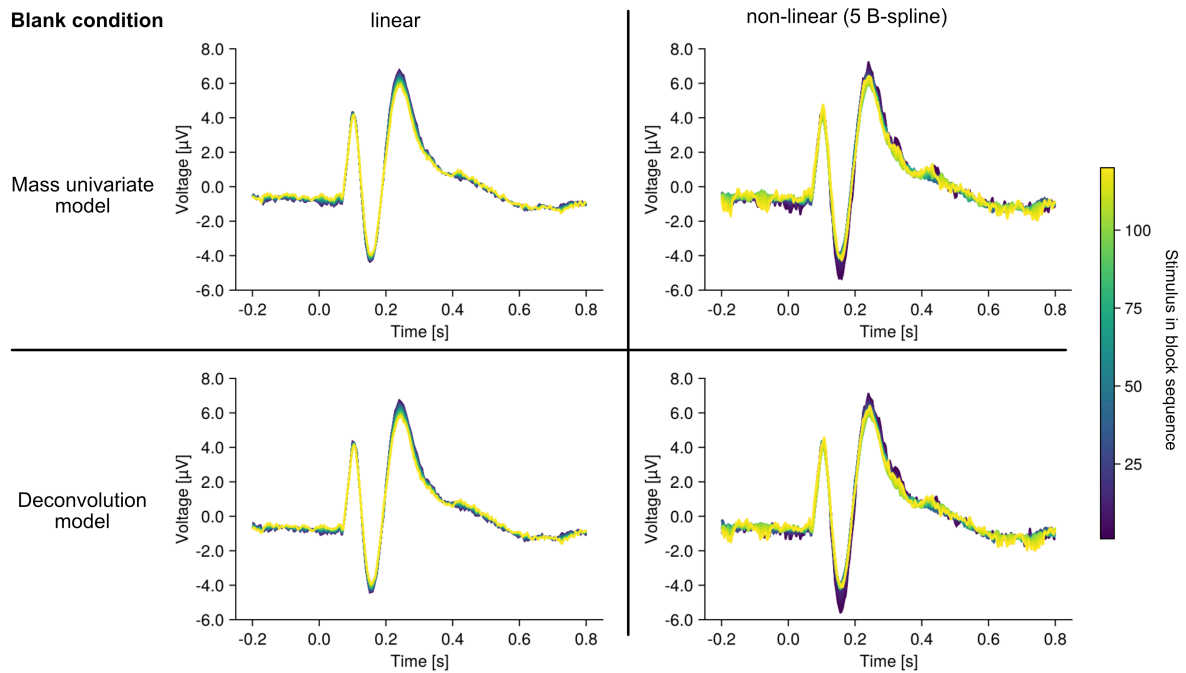


Figure 4.7 – Grand averaged ERPs (PO8 channel) for the task with varying inter-stimulus intervals (*blank* condition) modelled at different points in stimulus sequence of a trial block, indicated by the color bar on the right. The figure contains different modelling approaches (mass univariate, deconvolution), as well as linear and non-linear (5 B-spline) models.

For the *blank* condition trials, where log-uniformly distributed ISIs ranging from 800 to 2500ms were between the 120 face images in a block, the habituation effect seemed less pronounced. Only the non-linear models showed a decreasing effect within the stimulus sequence on the N170 component peak amplitude. The rest of the ERP seemed unaffected by the covariate.

/5

Discussion

The final chapter of this thesis consists of three subsections. Firstly the results of EEG analysis will be interpreted and set into the context of existing research. This also includes an assessment of the goals of this study and how the identified research gaps were filled. Then a section will discuss the limitations regarding the study design and evaluation, in order to address potential shortcomings of the research. Finally the scientific contribution of the work will be discussed, as well an outlook given, regarding potential implications and how further research could expand on the work in this thesis.

5.1 Interpretation

5.1.1 Inter-stimulus interval (ISI) effect

The results of the ISI effect showed that the amplitudes of all three investigated ERP components — P100, N170 and P300 — were reduced significantly in the RSVP task (*no-blank* condition) compared to the task with varying ISIs.

Regarding the P100 and N170 components, research on the isolated effect of ISIs is very sparse. One explanation for the attenuation effect on the peak

amplitudes could be the shorter refractory period - which describes the necessary recovery period of the neural circuits before they are able to fully fire again - in the *no-blank*. For the N170 component one study showed that recovery of the response occurred after 1400 to 1800 ms for faces as visual stimuli (Li et al., 2019). Further research revealed an adaptation effect - which describes the phenomenon that neural responses become smaller when the exposure to stimuli is repeated (Grill-Spector et al., 2006) - for faces as visual stimuli (Cao et al., 2015; Eimer et al., 2010; 2011). This effect was also observed in the analysis of this study as described in Chapter 4.3. However there are indications that in order to observe this adaptation effect for the N170 component short ISIs are necessary (Harris & Nakayama, 2007; Kuehl & others, 2013). This presumption also aligns with the findings of this work, since for the *blank* condition the adaptation effects regarding the N170 component were much smaller compared to the *no-blank* condition.

In contrast to the other two components several studies already investigated the effect of ISIs on the P300 component. An older work by Polich (1990) concluded that the P300 amplitude depends on the amount processing resources available when the component is produced for auditory stimuli. Further he argued that when auditory stimulus events occur infrequently - i.e. with longer ISIs - the neural system generating the P300 component can recover more fully and therefore larger component amplitudes are produced. The results of our study align with the assumption of Polich, showing a drastic decrease of P300 peak amplitude in the *no-blank* condition compared to the *blank* condition, which could be associated to the lack of recovery time for the neural system. While another study by Gonsalvez & Polich (2002) also confirmed this effect, it also concluded that the P300 is even more determined by the target-to-target interval in both auditory and visual stimuli conditions. However, since the study at hand doesn't differentiate between target and distractor stimuli, this effect can most likely be neglected in our case.

5.1.2 Combination of overlap correction & duration modelling

In the task with varying ISIs (*blank* condition), there were almost no differences between the mass univariate and deconvolution models regarding the marginal effects of the stimulus duration (Figure 4.3). This result was to be expected and also desired, since the ISIs reduced the overlap between ERPs and therefore the deconvolution model should produce similar results to the mass univariate model.

Now for the RSVP task (*no-blank* condition) the mass univariate and deconvolution models produced significantly different estimations regarding the stimulus duration effect. Neither the linear nor the non-linear mass univariate model estimated a duration effect for the late stage of the P300 component whereas both deconvolution models, linear and non-linear, consistently did. Further the non-linear mass univariate model produced a significant peak for very short durations right before the P300 component (Figure 4.2). We assume this peak is most likely caused by an overlap of ERP waveforms for very short stimulus durations, rather than an actual effect. For the linear mass univariate model this suspicious effect is also visible, however due to the linear confinement of the effect, it is most likely less pronounced in this model. For both of the deconvolution models there is no peak before the P300 component for any stimulus durations which seems to be a much more reasonable estimate, further affirming the applicability of the approach.

These qualitative observations coincide with the results by Skukies et al. (2025) from a simulated experiment which concluded that overlap correction and duration modelling can be applied together. The authors hereby simulated a duration effect using hanning functions as ‘ground truths’ which were then scaled according to a duration value. Additionally overlap was introduced between the functions. What they found was that of the models using overlap correction, the ones modelling the duration effect with splines (Chapter 2.2.4) performed best. This finding indicates that overlap correction and duration modelling can be used concurrently. Further they even found that in cases where overlap, but no duration effect was simulated, model performance only slightly decreased if the duration effect was modelled nevertheless. Therefore they argued that drawbacks of not considering a duration effect might be much bigger than incorporating it into the model, also in cases where overlap exists.

5.1.3 Stimulus duration effect

After we concluded in the previous section that the combination of deconvolution-based overlap correction and modelling of stimulus duration is viable - or at least for the study at hand - we will focus on the deconvolution models regarding the discussion of the RSVP task (*no-blank* condition), since they seemed to produce the most reasonable results. For the *blank* condition there was no difference between the mass univariate and deconvolution models which is why we will discuss both in unison.

As described in Chapter 4.2, for the RSVP task (*no-blank* condition) both the linear and non linear deconvolution model estimated a significant positively correlated stimulus duration effect for the later stage of the P300 component (see Figure 4.5 *no-blank* condition, 380ms latency for non-linear effect). Besides that no significant effect on the peaks of relevant ERP components could be detected statistically or by visual inspection of the ERPs.

A similar picture can be seen for the task with varying ISIs (*blank* condition), where also only the non-linear mass univariate and non-linear deconvolution models estimated a significant effect later stage of the P300 component according to the Hotelling's T^2 test. However, in the *blank* condition even this supposedly significant effect showed to have a non-monotonous shape (see Figure 4.5 *blank* condition, 380 ms latency), differing strongly from the *no-blank* condition effect. Therefore the actual correctness of the observed effect remains questionable. ERP components peaks remained unaffected by the stimulus durations in all models, like in the *no-blank* condition.

Overall previous research on the effect of stimulus duration on ERPs particularly for faces as visual stimuli is very sparse. The most relevant related work by Tanskanen et al. (2007) showed in a MEG study a positive correlation between stimulus duration and the amplitudes of the M170 and M300 cortical responses for faces as visual stimuli. Since the M170 and M300 are the MEG equivalents to the N170 and P300 ERP components, these effects could not reproduced in this study for the N170, neither in the *blank* nor *no-blank* task condition. Regarding the P300 component an effect was detected, as mentioned before, however only in the later stage of the component and not at the peak, as in the study Tanskanen et al. (2007). These differences in results are quite surprising, but could potentially be attributed to the higher task relevance of the stimulus in the mentioned study, since participants occasionally had to report on which face they saw. On the other hand in the study analyzed in this work the face images were completely irrelevant for the task - which was to recognize a red dot - possibly being an explanation for a lesser effect on the P300 components amplitude, since the component is modulated by task relevance and attention (Polich, 2007).

5.2 Limitations

Several potential limitations of the present study should be discussed. First, the use of baseline correction in a task involving rapid stimulus presentation

raises methodological concerns. In RSVP paradigms, successive stimuli are temporally close and neural responses strongly overlap. Conventional baseline correction hereby may introduce bias when the baseline window contains residual activity from preceding events. This can lead to systematic distortions of component amplitudes, also in cases where ISIs vary or are very short. Although baseline correction is standard in traditional ERP analysis, its appropriateness in the context of overlapping responses remains debated, and the present choice may have affected the observed amplitude differences between conditions (Alday, 2019).

Second, aspects of the experimental design may cause further constraints. The use of a log-uniform distribution for the stimulus duration resulted in an imbalanced distribution of trials across possible stimulus durations. This circumstance might possibly impair the ability of the regression models to correctly estimate the duration effect for longer stimulus durations.

5.3 Conclusion & Future Scope

The work in this thesis showed that ISIs can have a significant effect on ERPs regarding the P100, N170 and P300 components in a task with faces as visual stimuli (Figure 4.1). Due to this significant difference between the task conditions the effect of ISIs duration could be of interest for future studies. The effect could hereby be modelled continuously with slope rERPs, similar to the stimulus duration effect in this work.

An additional analysis further showed a strong adaptation effect regarding the N170 component during a sequence of faces in a stimulus block, especially in the RSVP paradigm (Figure 4.6). This effect could also be of interest for further research or at least should to be considered in future experiments, especially regarding RSVP tasks.

Moreover, the applicability of deconvolution-based overlap correction in combination with modelling of stimulus duration was further consolidated, supporting the hypothesis by Skukies et al. (2025) from simulated data, with the results from a real experiment.

Nevertheless, since results from previous studies regarding the stimulus duration effect with faces as visual stimuli couldn't be reproduced, further research with an adapted study design could be conducted on this matter. Potential

adaptions could hereby be the choice of another distribution for the stimulus durations as well as ISIs - potentially a uniform distribution so that every duration has an equal amount of trials. This however would lead to a monotonically increasing hazard rate (Herbst et al., 2018; Nobre et al., 2007) over time potentially causing other drawbacks which would need to be critically evaluated beforehand given the context of the study. Another idea would be to alter the experimental design in order to make the visual stimulus (face image) more relevant for the task. This could potentially enable a better assessment of the effect on the endogenous P300 component, since it is known to be modulated by attention and task relevance (Polich, 2007).

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Zusammenfassung

Vorherige Untersuchungen haben gezeigt, dass die Dauer eines Ereignisses ein Störfaktor in der EEG-Forschung sein kann. Daher wurde in dieser Studie der Einfluss der Stimulusdauer auf Ereigniskorrelierte Potenziale (EKP) unter Verwendung von Gesichtern als visuelle Stimuli untersucht. Die Studie, welche von Martin Geiger im Rahmen seiner Masterarbeit durchgeführt wurde, bestand aus einer Aufgabe wo Stimuli in direkter Abfolge (RSVP Paradigma), mit logarithmisch gleichmäßig verteilten Reizdauern (100–1500 ms) und ohne Interstimulusintervalle (ISIs) präsentiert wurden. Weiterhin mussten die Probanden eine Kontrollaufgabe durchführen, die von der Konzeption beinahe identisch war, allerdings variierende ISIs (800–2500 ms) hatte. Die in der Studie gesammelten Daten wurden in MATLAB, vorverarbeitet und mittels der julia toolbox *Unfold.jl*, die das regressionsbasierte *rERP* framework implementiert analysiert. Die Ergebnisse zeigten keinen signifikanten Einfluss der Stimulusdauer auf die ERP-Komponenten P100, N170 und P300, jedoch einen signifikanten positiv korrelierten Effekt auf die späte Phase der P300 Komponente (Latenz ~300–500 ms). Darüber hinaus wurde ein signifikanter Einfluss der ISIs auf die Spitzenamplituden der Komponenten P100, N170 und P300 nachgewiesen. Während einer zusätzlichen Analyse wurde ein Adaptionseffekt hinsichtlich der N170 Komponente während einer Sequenz visueller Stimuli beobachtet. Schließlich ergab eine qualitative Bewertung der Ergebnisse, dass eine Kombinierte Anwendung von dekonvolutionsbasierter Überlappungskorrektur und Stimulusdauer-Modellierung methodisch fundiert ist.

Eigenständigkeitserklärung

Hiermit erkläre ich, dass ich die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel (dazu zählen auch KI-basierte Anwendungen oder Werkzeuge) benutzt habe. Sämtliche wörtlichen oder sinngemäßen Übernahmen und Zitate sind kenntlich gemacht und nachgewiesen. Ich versichere, dass ich keine Hilfsmittel verwendet habe, deren Nutzung die Prüferin oder der Prüfer explizit ausgeschlossen hat.

Im Anhang „Usage of AI“ habe ich die von mir verwendeten KI-Tools dokumentiert. Mit Abgabe der vorliegenden Arbeit übernehme ich die Verantwortung für das eingereichte Gesamtprodukt. Ich verantworte damit auch jegliche KI-generierten Inhalte, die ich in meine Arbeit übernommen habe. Die Richtigkeit übernommener (KIgenerierter) Aussagen und Inhalte habe ich nach bestem Wissen und Gewissen geprüft.

Weder diese Arbeit noch wesentliche Teile daraus waren bisher Gegenstand eines anderen Prüfungsverfahrens. Ich habe diese Arbeit bisher weder teilweise noch vollständig veröffentlicht. Das elektronische Exemplar stimmt mit allen eingereichten Exemplaren überein.

Datum, Unterschrift des/der Studierenden

Usage of AI

ChatGPT and Gemini were used to generate some code snippets for basic programming tasks in the MATLAB preprocessing script, as well as the analysis scripts in julia.

Further the two models were used to find existing studies or scientific sources on topics, where research was very sparse. Cited sources were hereby always double checked, in order to ensure correctness and scientific integrity.

